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Subject: Programming with Java

ASSIGNMENT No : 3

TITLE: Method Overloading and Static Members

Problem Statement

Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

Learning Objectives

To implement method overloading and use static variables and methods in Java programs.

Programme

```
1  class Calculator {
2      static int calculationCount = 0;
3
4      // Addition of integers
5      int add(int a, int b) {
6          calculationCount++;
7          return a + b;
8      }
9
10     // Addition of decimals
11     double add(double a, double b) {
12         calculationCount++;
13         return a + b;
14     }
15
16     static void showCount() {
17         System.out.println("Total Calculations: " + calculationCount);
18     }
19
20     Run | Debug
21     public static void main(String[] args) {
22         Calculator c = new Calculator();
23
24         System.out.println("Integer Addition: " + c.add(a: 10, b: 20));
25         System.out.println("Decimal Addition: " + c.add(a: 10.5, b: 20.8));
26
27         Calculator.showCount();
28     }
29 }
```

```
J RestaurantDemo.java > RestaurantBilling
1  class RestaurantBilling {
2      static int totalOrders = 0;
3
4      // Dine-in bill
5      double calculateBill(double amount) {
6          totalOrders++;
7          return amount;
8      }
9
10     // Takeaway bill (with packing charge)
11     double calculateBill(double amount, double packingCharge) {
12         totalOrders++;
13         return amount + packingCharge;
14     }
15
16     // Delivery bill (with packing and delivery charge)
17     double calculateBill(double amount, double packingCharge, double deliveryCharge) {
18         totalOrders++;
19         return amount + packingCharge + deliveryCharge;
20     }
21
22     static void showTotalOrders() {
23         System.out.println("Total Orders: " + totalOrders);
24     }
25
26     Run | Debug
27     public static void main(String[] args) {
28         RestaurantBilling bill = new RestaurantBilling();
29
30         System.out.println("Dine-in Bill: ₹" + bill.calculateBill(amount: 500));
31         System.out.println("Takeaway Bill: ₹" + bill.calculateBill(amount: 500, packingCharge: 30));
32         System.out.println("Delivery Bill: ₹" + bill.calculateBill(amount: 500, packingCharge: 30, deliveryCharge: 50));
33
34         RestaurantBilling.showTotalOrders();
35     }
}
```

Output

```
Integer Addition: 30
Decimal Addition: 31.3
Total Calculations: 2
```

```
Dine-in Bill: ₹500.0
Takeaway Bill: ₹530.0
Delivery Bill: ₹580.0
Total Orders: 3
```

Conclusion